

An unjustified answer will be considered as false

It is imperative to use the provided squared sheets for the simplified diagrams

We consider an implementation called P9. P9 has the same instruction set as Mips-32. It is a 9-stage pipelined implementation : **IFC1**, **IFC2**, **IFC3**, **DEC**, **EXE**, **MEM1**, **MEM2**, **MEM3**, **WBK**.

In this implementation, memory accesses are performed in three cycles. The **DEC** cycle decodes the instruction and prepares the operands. The **EXE** cycle performs the operation and **calculates the address of the next instruction**. The **WBK** cycle is similar to the Mips-32 **WBK** cycle.

- 1- Does this 9-stage pipeline organization have any impact on the compiler ?
- 2- In the Mips-32 implementation studied in class, there are data dependencies of order 1, 2 and 3. In the P9 realization, up to what order can data dependencies be observed ?
- 3- In Mips-32 8 bypasses are used to limit the degradation of performance due to data dependencies. What are the bypasses in the P9 implementation ? For each bypass, give an example of a sequence of instructions that illustrates the need for this bypass.

We would like to study the performance of execution of a sorting function on this implementation. Two different versions of the same function will be considered.

Consider the **MaxTab** function, used in the maximum sort algorithm to sort an array of integers. It takes an array of integers **t** as input. Scanning the array, it identifies the position of the maximal element in the array. Then, it moves this element to the end of the array.

<pre>int *MaxTab (int *t, unsigned int size) { unsigned int i = 0; unsigned int max_idx = 0; int max = 0; for (i=0 ; i!=size ; i++) if (t[max_idx] < t[i]) max_idx = i;</pre>	<pre>max = t [max_idx]; t [max_idx] = t [size-1]; t [size-1] = max; return (t); }</pre>
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A first (non-pipelined) compiler has produced a first version of the assembly code for the main loop of this function. It is assumed that, at loop entry, register R4 contains the address of array **t**, R5 the end address of array **t** (the address immediately following that of the last cell in the array) and R6 the address of the cell in array **t** with index **max_idx** (the variable **max_idx**). R7 will contain the value of **t[max_idx]**.

Version 1 :

```
_For : Lw      r7 , 0 (r6 )      ; t[max_idx]
       Lw      r8 , 0 (r4 )      ; t[i]
       Slt     r9 , r7 , r8      ; t[max_idx] < t[i] ?
       Beq     r9 , r0 , _Endif
       Or      r6 , r4 , r0
_Endif: Addiu   r4 , r4 , 4
       Bne     r4 , r5 , _For
```

- 4- Modify the code so that it can be executed on P9. Analyze the execution of the loop using a simplified diagram in the case where branch fails.
- 5- How many cycles are required to execute one iteration of the loop in the case of a failed branch and in the case of a successful branch ? Calculate the CPI and CPI-useful, assuming that in 30% of cases the branch fails.
- 6- Optimize the code by modifying the order of instructions to avoid lost cycles as much as possible (explain your approach). How many cycles are needed to perform an iteration ? You don't need to draw a diagram, but simply show on the code the lost cycles after optimization.
- 7- To improve execution performance, we decide to unroll the code once (processing 2 elements at each iteration). Propose an optimized code. How many cycles are needed to process an element ? You don't need to draw a diagram, but simply show on the code the lost cycles after optimization.

More recent versions of the Mips architecture provides the **Movn** (Move if Not Zero) instruction. This R-format instruction transfers the contents of register Rs into register Rd if the value of register Rt is not zero. This instruction is executed in the pipeline as a computation instruction (the condition is evaluated in the **EXE** cycle). Using this instruction, we can write a second version of the loop.

Version 2 :

```

_For : Lw      r7 , 0 (r6 )    ; t[max_idx]
       Lw      r8 , 0 (r4 )    ; t[i]
       Slt     r9 , r7 , r8    ; t[max_idx] < t[i] ?
       Movn    r6 , r4 , r9
       Addiu   r4 , r4 , 4
       Bne     r4 , r5 , _For

```

- 8- After adapting the code of version 2 to P9, how many cycles are required to execute one iteration of the loop ? Calculate the CPI-useful. It is not necessary to draw a diagram, but simply indicate the lost cycles on the code.
- 9- Optimize the loop code by modifying the order of instructions to avoid lost cycles as much as possible. How many cycles are needed to perform an iteration ? You don't need to draw a diagram, but simply indicate the lost cycles on the code after optimization.
- 10- To improve execution performance, we decide to unroll the loop code once (processing 2 elements at each iteration). Propose an optimized code. How many cycles are needed to process an element ? You don't need to draw a diagram, but simply indicate the lost cycles on the code after optimization.